Central Limit Theorem

MDragon Data Tools - Study Guide

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1. What is the Central Limit Theorem (CLT)?

The **Central Limit Theorem** states that:

> *When you take sufficiently large random samples from any population (with any shape), the distribution of sample means will be approximately normally distributed, regardless of the population's original distribution.*

In Simple Terms

- Take many samples from a population
- Calculate the mean of each sample
- Plot all those sample means

- The distribution will look like a bell curve (normal distribution)
- **This works even if the original population is NOT normal!**

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2. Why It Matters

The CLT is one of the most important theorems in statistics because:

1. **Enables inference** - We can make conclusions about populations using normal distribution theory
2. **Works with any distribution** - Original population can be skewed, uniform, bimodal, etc.
3. **Foundation for hypothesis testing** - Allows us to test claims about population means
4. **Basis for confidence intervals** - Enables us to estimate population parameters
5. **Justifies using Z and T tests** - Even when population isn't normal

Real-World Applications

- Political polling and surveys
- Quality control in manufacturing
- Medical research and clinical trials
- Financial risk assessment
- A/B testing in business

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3. Key Concepts

Sampling Distribution of Sample Means

- **Population**: All individuals of interest
- **Sample**: Subset taken from population
- **Sample mean ($\bar{x}$)**: Average of sample values
- **Sampling distribution**: Distribution of ALL possible sample means

Properties of the Sampling Distribution

1. Mean of sampling distribution:

```
μₓ̄ = μ (equals population mean)
```

2. Standard deviation of sampling distribution (Standard Error):

```
σₓ̄ = σ / √n

where:
σ = population standard deviation
n = sample size
```

3. Shape:

- Approximately normal (if $n \geq 30$ or population is normal)
- More normal as n increases
- Center closer to μ as n increases
- Spread decreases as n increases

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4. Requirements and Conditions

For CLT to Apply:

1. Sample Size Rule of Thumb:

- **$n \geq 30$**: CLT applies for most distributions
- **$n < 30$**: CLT still works if population is approximately normal
- **Severely skewed**: May need $n > 30$ (sometimes $n \geq 40$)

2. Independence:

- Random sampling

- If sampling without replacement: $n \leq 0.05N$ (sample < 5% of population)

3. Sample vs Population:

- Each sample is a random, independent observation
- Samples drawn from the same population

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5. Step-by-Step Examples

Example 1: Basic CLT Application

📝 **Problem:** A population of test scores has $\mu = 70$ and $\sigma = 15$. If we take samples of size $n = 36$, describe the sampling distribution of $\bar{x}$.

💡 **Solution:**

1. **Check if CLT applies:**
   - $n = 36 \geq 30$ ✓
   - Random sampling assumed ✓
2. **Characteristics of sampling distribution:**

```
Shape: Approximately normal (due to CLT)

Mean: μₓ̄ = μ = 70

Standard Error: σₓ̄ = σ / √n
                    = 15 / √36
                    = 15 / 6
                    = 2.5
```

✅ **Answer:**

- The sampling distribution of $\bar{x}$ is approximately normal
- Center: $\mu_{\bar{x}} = 70$
- Spread: $\sigma_{\bar{x}} = 2.5$

Interpretation: Sample means will typically be within 2.5 points of the true mean (70).

Example 2: Probability About Sample Mean

📝 **Problem:** Using the scenario from Example 1, what is the probability that a random sample of 36 scores has a mean greater than 75?

💡 **Solution:**

1. **We know from Example 1:**
   - $\mu_{\bar{x}} = 70$
   - $\sigma_{\bar{x}} = 2.5$
   - Distribution is approximately normal
2. **Calculate z-score:**

```
z = (x̄ - μₓ̄) / σₓ̄
  = (75 - 70) / 2.5
  = 5 / 2.5
  = 2
```

3. **Find probability:**

```
P(x̄ > 75) = P(Z > 2)
           = 1 - P(Z < 2)
           = 1 - 0.9772
           = 0.0228
```

✅ **Answer:** 0.0228 or 2.28%

Interpretation: Only about 2.3% of samples of size 36 will have means above 75.

Example 3: Effect of Sample Size

📝 **Problem:** A population has $\mu = 100$ and $\sigma = 20$. Compare the standard error for:

a) n = 25

b) n = 100

💡 **Solution:**

a) n = 25:

$$\sigma_{\bar{x}} = 20 / \sqrt{25} = 20 / 5 = 4$$

b) n = 100:

$$\sigma_{\bar{x}} = 20 / \sqrt{100} = 20 / 10 = 2$$

✅ **Answer:**

- n = 25: SE = 4
- n = 100: SE = 2

Interpretation:

- Quadrupling the sample size (25→100) cuts the standard error in half
- Larger samples give more precise estimates (less variability)
- To cut SE in half, you need 4× the sample size

Example 4: Non-Normal Population

📝 **Problem:** A population is uniformly distributed (flat, not normal) with $\mu = 50$ and $\sigma = 10$. If we take samples of n = 40, can we use the CLT? What is the probability that $\bar{x} < 48$?

💡 **Solution:**

1. **Check if CLT applies:**

```
n = 40 ≥ 30 ✓
Even though population is uniform (not normal),
CLT says sampling distribution will be approximately normal
```

2. **Calculate parameters:**

```
μₓ̄ = 50
σₓ̄ = 10 / √40 = 10 / 6.325 = 1.581
```

3. **Calculate z-score:**

```
z = (48 - 50) / 1.581
  = -2 / 1.581
  = -1.265
```

4. **Find probability:**

```
P(x̄ < 48) = P(Z < -1.265) ≈ 0.1029
```

✅ **Answer:** 0.1029 or 10.29%

Key Point: Even though the population is uniform (NOT normal), we can use normal distribution methods for the sample mean because $n \geq 30$!

Example 5: Finding Sample Size

📝 **Problem:** A population has $\sigma = 16$. What sample size is needed so that the standard error is at most 2?

💡 **Solution:**

1. **Set up inequality:**

```
σₓ̄ ≤ 2
σ / √n ≤ 2
16 / √n ≤ 2
```

2. **Solve for n:**

```
16 ≤ 2√n
8 ≤ √n
64 ≤ n
```

✅ **Answer:** Need at least n = 64

Interpretation: To achieve a standard error of 2 or less, we need samples of at least 64 observations.

Example 6: Range of Sample Means

📝 **Problem:** Heights have μ = 68 inches and σ = 4 inches. For samples of n = 64, what range contains the middle 95% of sample means?

💡 **Solution:**

1. **Calculate standard error:**

```
σₓ̄ = 4 / √64 = 4 / 8 = 0.5
```

2. **Use 95% rule (within 2 standard errors):**

```
μₓ̄ ± 2σₓ̄ = 68 ± 2(0.5)
           = 68 ± 1
           = (67, 69)
```

✅ **Answer:** (67, 69) inches

Interpretation: 95% of sample means (from samples of size 64) will fall between 67 and 69 inches.

Example 7: Comparing Individual vs Sample Mean

📝 **Problem:** IQ scores have $\mu = 100$ and $\sigma = 15$.

a) What percentage of individuals have IQ > 115?

b) For samples of $n = 25$, what percentage of sample means are > 115?

💡 **Solution:**

a) Individual scores:

```
z = (115 - 100) / 15 = 1
P(X > 115) = P(Z > 1) = 1 - 0.8413 = 0.1587 or 15.87%
```

b) Sample means (n = 25):

```
σₓ̄ = 15 / √25 = 3
z = (115 - 100) / 3 = 5
P(x̄ > 115) = P(Z > 5) ≈ 0.0000003 (virtually 0%)
```

✅ **Answer:**

- Individual: 15.87%
- Sample mean: ~0%

Interpretation: While individual scores above 115 are common (about 16%), getting a sample average above 115 is extremely rare. Sample means are much less variable than individual scores.

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6. Practice Problems

Problem 1

A population has $\mu = 200$ and $\sigma = 30$. For samples of $n = 36$:

a) What is the mean of the sampling distribution?

b) What is the standard error?

c) Describe the shape of the sampling distribution.

Problem 2

SAT scores have $\mu = 1000$ and $\sigma = 200$. What is the probability that a random sample of 64 students has a mean score above 1050?

Problem 3

A population standard deviation is $\sigma = 12$. What sample size is needed for the standard error to be 2 or less?

Problem 4

Waiting times are uniformly distributed with $\mu = 10$ minutes and $\sigma = 5$ minutes. For samples of $n = 50$:
a) Can we use CLT?
b) What is $P(\bar{x} < 9)$?

Problem 5

Compare the standard error for $\sigma = 20$ with:
a) $n = 16$
b) $n = 64$
What is the effect of quadrupling the sample size?

Answers to Practice Problems

Problem 1

```
a) μₓ̄ = μ = 200
b) σₓ̄ = 30 / √36 = 5
c) Approximately normal (n ≥ 30)
```

Problem 2

$\sigma_{\bar{x}} = 200 / \sqrt{64} = 25$

$z = (1050 - 1000) / 25 = 2$

$P(\bar{x} > 1050) = P(Z > 2) = 0.0228$ or 2.28%

Problem 3

$\sigma_{\bar{x}} = \sigma / \sqrt{n} \leq 2$

$12 / \sqrt{n} \leq 2$

$\sqrt{n} \geq 6$

$n \geq 36$

Problem 4

a) Yes, $n = 50 \geq 30$

b) $\sigma_{\bar{x}} = 5 / \sqrt{50} = 0.707$

   $z = (9 - 10) / 0.707 = -1.414$

   $P(\bar{x} < 9) = P(Z < -1.414) \approx 0.0787$ or 7.87%

Problem 5

a) $n = 16$: $\sigma_{\bar{x}} = 20 / 4 = 5$

b) $n = 64$: $\sigma_{\bar{x}} = 20 / 8 = 2.5$

Effect: SE is cut in half (50% reduction)

Quick Reference

Sampling Distribution of x̄

```
Mean: μₓ̄ = μ
Standard Error: σₓ̄ = σ / √n
Shape: Approximately normal if n ≥ 30
```

Z-Score for Sample Mean

```
z = (x̄ - μ) / (σ / √n)
```

Key Insights

- Larger n → Smaller $\sigma_{\bar{x}}$ → More precise
- To cut SE in half → Need 4× the sample
- CLT works for ANY population distribution (if n large enough)

The CLT is the foundation for most statistical inference - master it to understand hypothesis testing and confidence intervals!
